

Q1: Express the following hexadecimal to decimal using IEEE 754 floating point Representation.

⇒ 0x40f00000

0100 0000 1111 0000 0000 0000 0000 0000
 1bit 8bit 23bit

$$(-1)^S * (1 + \text{Significand}) * (2^{\text{Exponent} - \text{Bias}})$$

$$(-1)^0 * (1 + 0.875) * (2^{129 - 127})$$

7.5

كيفية الحساب

* Significand : $2^{-1} * 1 + 2^{-2} * 1 + 2^{-3} * 2 = 0.875$

⇒ 0xc1400000

1100 0001 0100 0000 0000 0000 0000 0000
 1bit 8bit 23bit

$$(-1)^S * (1 + \text{Significand}) * (2^{\text{Exponent} - \text{Bias}})$$

$$(-1)^1 * (1 + 0.5) * (2^{130 - 127})$$

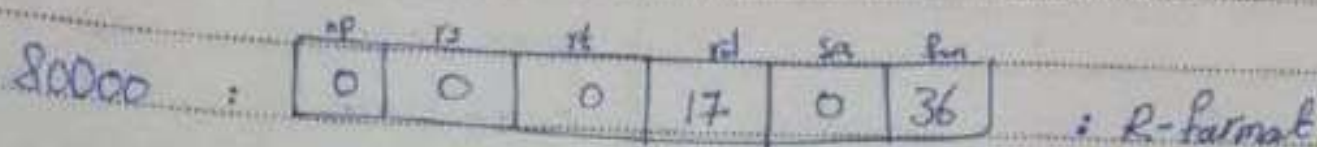
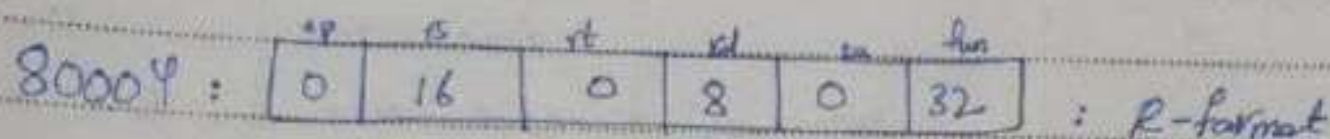
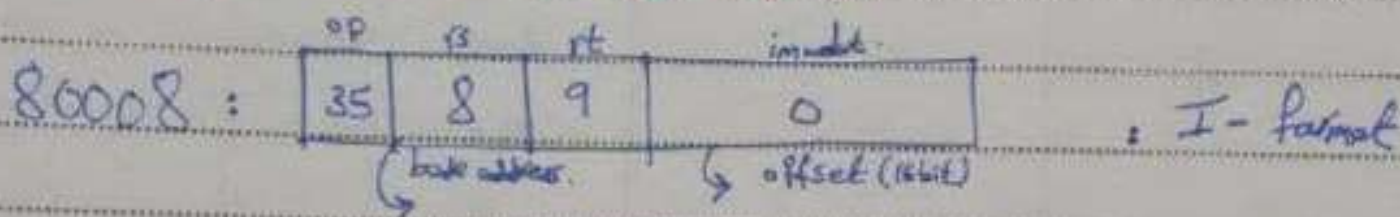
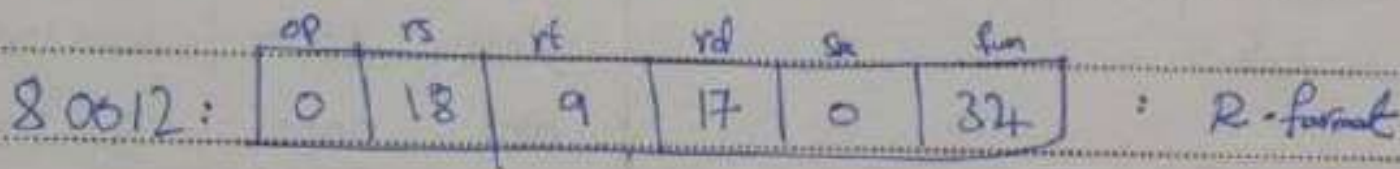
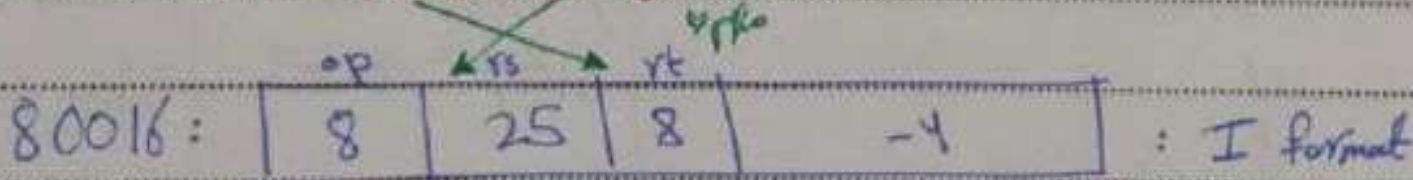
-12

Subject:

Q2/ Complete (0101) / (0011) 8

Iteration	Divisor (D)	Remainder (R)	Remark
0		$\begin{array}{r} 1000 \ 0101 \\ 0000 \ 1010 \end{array}$	initial state. $R = R \ll 1$
1	0011	$\begin{array}{r} 1101 \ 1010 \\ 0000 \ 1010 \\ \hline 0001 \ 0100 \end{array}$	$\text{left}(R) = \text{left}(R) - D$ undo $R = R \ll 1, R_0 = 0$
2		$\begin{array}{r} 1110 \ 0100 \\ 0001 \ 0100 \\ \hline 0010 \ 1000 \end{array}$	$\text{left}(R) = \text{left}(R) - D$ undo $R = R \ll 1, R_0 = 0$
3		$\begin{array}{r} 111 \ 1000 \\ 0010 \ 1000 \\ \hline 0101 \ 0000 \end{array}$	$\text{left}(R) = \text{left}(R) - D$ undo $R = R \ll 1, R_0 = 0$
4		$\begin{array}{r} 0101 \ 0000 \\ 0101 \ 0001 \\ \hline 0000 \end{array}$	$\text{left}(R) = \text{left}(R) - D$ $R = R \ll 1$ $R_0 = 1$
Extra: :		$\begin{array}{cc} R & Q \\ \boxed{0010} & \boxed{0001} \end{array}$	$R = R \gg 1$

Q3

and \$s₁, \$zero, \$zero② add \$t₀, \$s₀, \$zero③ loop: lw \$t₁, 0(\$t₀)④ sub \$s₁, \$s₂, \$t₁⑤ addi \$t₀, \$t₉, -4

الموقع الي بدي اوصله = الموقع الي
انا فيه (PC) + 4 + 4X بعدها
بحسب X يلي هي offset المطلوبة

⑥ bne \$t₁, \$zero, loop

Subject:

Q4]

from 0 to 9 int Sum = 0;

int i;

for (i = 0; i < 10; i = i + 1) Sum = Sum + i

\$S0 = i code, \$S1 = Sum ?

Sol :

addi \$S1, \$Zero, 0 Sum = 0 = S1

addi \$S0, \$Zero, 0 # i = 0 = S0

addi \$t0, \$Zero, 10 # t0 = 10

loop: add \$S1, \$S1, \$S0 # Sum = Sum + i

addi \$S0, \$S0, 1 # i + 1

beg \$S0, \$t0, Exit

J loop

Exit

- explain operand Addressing Modes

- ① Register addressing $\rightsquigarrow$ ex: add, sub
- ② Base addressing $\rightsquigarrow$ ex: load, store
- ③ Immediate addressing $\rightsquigarrow$ ex: addi
- ④ pc - relative addressing $\rightsquigarrow$ ex: branch.
- ⑤ pseudo - direct addressing $\rightsquigarrow$ ex: jum

هاد كان أول سؤال ب امتحان الميدع
الفصل الثاني الدكتور زاد عدد الاسئلة وقلل
العلامات عليهم

A	3	معمارية الحاسوب وتنظيمه	
~ Dania Khalid alkhawwar ●●●			
20 / 20 / 0	30 / 30	/ 0	معمارية الحاسوب وتنظيمه